VOICE-BASED TRANSPORT ENQUIRY SYSTEM

A PROJECT REPORT

Submitted by

THIRUPPATHI P

in partial fulfilment for the award of the course

CGB1221 – DATABASE MANAGEMENT SYSTEMS

IN

DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
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CHENNAI 600 025

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PROJECT FINAL DOCUMENT

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BONAFIDE CERTIFICATE

Certified that this project report titled **"VOICE-BASED TRANSPORT ENQUIRY SYSTEM"** is the bonafide work of **THIRUPPATHI P (8115U24AM060)** who carried out the work under my supervision.

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DECLARATION BY THE CANDIDATE

I declare that to the best of my knowledge the work reported here in has been composed solely by myself and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva-Voice held at K. Ramakrishnan College of Engineering on __________

SIGNATURE OF THE CANDIDATE

ACKNOWLEDGEMENT

I thank the almighty GOD, without whom it would not have been possible for me to complete my project.

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I wish to convey my sincere thanks to **Dr.B.KIRAN BALA, M.E., M.B.A., Ph.D.,** Head of the Department, Artificial Intelligence and Machine Learning for giving me constant encouragement and advice throughout the course.

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DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

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M2 To foster Experiential learning equips students with engineering skills to tackle real-world problems.

M3 To promote collaborative innovation in AI, machine learning, and related research and development with industries.

M4 To provide an enjoyable environment for pursuing excellence while upholding strong personal and professional values and ethics.

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PO5 Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

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PO7 Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO8 Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings

PO9 Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10 Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one’s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11 Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

The **Voice-Based Transport Enquiry System** is a web-based application designed to enable users to search for transport details efficiently using both voice and manual inputs. The system allows users to enquire about available transport services between locations by speaking their queries or entering them manually through an interactive interface. It uses Python and Streamlit for the frontend, with MySQL as the backend database for managing transport routes, vehicle information, and user data. Speech recognition is integrated to convert voice input into text, which is then processed to extract source and destination cities for database queries. All data is stored in a normalized relational database to ensure consistency, integrity, and efficient retrieval. User authentication is implemented to provide secure access, with role-based control for admin and users. The admin module allows adding, updating, and deleting transport records to maintain the database effectively. CRUD operations are used throughout the system for dynamic data handling. The application also includes route visualization with distance and time estimation to enhance user experience. Basic analytics features are included to display transport statistics and usage insights. The system is lightweight, scalable, and suitable for academic and real-time applications. The project demonstrates the practical use of DBMS concepts integrated with voice technology to build an intelligent transport enquiry solution.

ABSTRACT WITH POs AND PSOs MAPPING

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Note: 1- Low, 2-Medium, 3- High

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LIST OF ABBREVIATIONS

S. NO.	ACRONYM	ABBREVIATION
1	API	Application Programming Interface
2	CRUD	Create, Read, Update, Delete
3	DBMS	Database Management System
4	GPS	Global Positioning System
5	HTTP	HyperText Transfer Protocol
6	OSRM	Open-Source Routing Machine
7	SQL	Structured Query Language
8	STT	Speech-to-Text
9	UI	User Interface
10	UX	User Experience

LIST OF FIGURES

CHAPTER 1
INTRODUCTION

1.1 OBJECTIVE

The primary objective of the **Voice-Based Transport Enquiry System** project is to design and develop a user-centric application that enables individuals to easily access transport information using voice and manual inputs. In the current digital age, fast and convenient access to travel details has become essential, and this system aims to provide an efficient solution by allowing users to speak their queries and receive instant results. The platform offers a structured environment for searching, managing, and visualizing transport data, making it suitable for real-time usage and improving overall user experience. The system is developed with the goal of implementing all essential CRUD (Create, Read, Update, Delete) functionalities for transport data and user interactions. It ensures that users can search for routes between locations, view transport details such as vehicle type, departure time, and fare, and interact with the system through both voice recognition and manual input. Security is also an important aspect, and the system incorporates authentication mechanisms to ensure controlled access to administrative features and protect data integrity. Overall, the project demonstrates an efficient and intelligent transport enquiry solution by integrating DBMS concepts with modern voice-enabled technology.

1.2 SCOPE OF THE PROJECT

The **Voice-Based Transport Enquiry System** is designed as a smart and interactive web-based application that enables users to access transport information efficiently using both voice and manual inputs. The scope of the project is broad and adaptable, making it suitable for various users such as daily commuters, students, travelers, and administrators who require quick and reliable transport enquiry services.

At its core, the system allows users to register, log in, and perform transport searches through a secure and user-friendly interface developed using Streamlit. Authenticated users can interact with the system either by speaking their queries through a microphone or by manually selecting source and destination cities. The voice input is processed using speech recognition techniques, where spoken words are converted into text and analyzed to extract relevant city names. Based on the input, the system retrieves matching transport data from the database and displays it in structured formats such as tables, smart cards, and downloadable files. The scope of the project includes multiple functional modules such as the User Management Module for authentication and account handling, the Voice Enquiry Module for processing spoken queries, the Manual Enquiry Module for direct user input, the Transport Data Module for database operations, and the Admin Module for system management. It also includes an Analytics Module that processes enquiry logs to generate useful insights such as frequently searched routes and system usage patterns.

1.3 OVERVIEW OF THE SYSTEM

The Voice-Based Transport Enquiry System is a full-stack web application that enables users to search, manage, and retrieve transport information through an interactive interface integrated with a structured database. The system is built using Python with Streamlit for the frontend, along with backend processing and a relational database management system (MySQL), providing a seamless and efficient transport enquiry experience. The core functionalities of the system include: New users can register with a username and password, which are securely stored using hashing techniques. Once authenticated, users gain access to system features such as voice and manual enquiry modules. Users can search for transport details by either speaking their query or selecting source and destination manually. The system processes the input, retrieves relevant transport records including vehicle type, fare, and timing, and displays the results in structured formats. Transport results are displayed using tables and smart cards, allowing users to easily compare options. The system also provides downloadable data and supports filtering based on transport type and fare range. The application integrates speech recognition to convert voice input into text and extract city names. It processes user inputs, executes SQL queries, and returns relevant results for display. The database maintains structured tables for users, transport records, routes, and enquiry logs. Relationships between entities are maintained using keys, and the schema is designed to ensure data consistency and efficient retrieval. The system implements security mechanisms such as password hashing, user authentication, role-based access control, and input validation to protect data and prevent unauthorized access. Overall, the Voice-Based Transport Enquiry System provides a comprehensive solution for accessing transport information. It combines voice technology, database management, and interactive visualization to deliver a modern, efficient, and scalable transport enquiry platform.

1.4 IMPORTANCE OF DBMS IN TRANSPORT SYSTEMS

A **Database Management System (DBMS)** is a core component of any data-driven application, and its importance is especially significant in systems like the Voice-Based Transport Enquiry System. The application handles continuous data interactions from multiple users, including transport records, user credentials, enquiry logs, routes, and timestamps. Managing this volume of structured data in a reliable, organized, and secure manner is made possible through the use of a DBMS. The primary role of the DBMS in this project is to provide a structured mechanism to store, retrieve, update, and manage transport-related data efficiently. By defining constraints and validation rules at the database level, the system prevents incorrect or incomplete data from being stored, thereby maintaining data quality. Sensitive information such as user passwords is securely stored using hashing techniques, and access control mechanisms ensure that only authorized users, such as administrators, can modify or delete critical data. In case of unexpected failures, the DBMS supports data persistence and recovery mechanisms, ensuring that important transport and user data is not lost. As the number of users and transport records increases, the DBMS can efficiently scale to handle larger datasets while maintaining performance through optimized queries and caching techniques. The system leverages database operations to support advanced features such as analytics, including tracking popular routes, enquiry trends, and usage patterns through aggregate queries. In summary, the DBMS acts as the backbone of the Voice-Based Transport Enquiry System, enabling smooth and reliable data operations. It ensures that all functionalities—from user authentication to transport search and analytics. Without a robust DBMS, the system would face challenges in maintaining data consistency and scalability.

1.5 TECHNOLOGIES USED

The Voice-Based Transport Enquiry System is developed using a combination of frontend, backend, and database technologies. Each layer of the application—user interface, processing logic, and data storage—is implemented using tools and libraries that are efficient, reliable, and scalable. These technologies work together to deliver an interactive, intelligent, and fully functional web-based transport enquiry system. The main technologies used in this project are categorized as follows: Streamlit is used to build the user interface of the application. It provides components such as buttons, forms, tables, tabs, and sidebars, allowing users to interact with the system easily. HTML/CSS (via Streamlit Styling) Basic styling and layout enhancements are applied to improve the visual appearance of the application. This includes formatting text, organizing sections, and ensuring a clean and user-friendly interface. The system includes dynamic elements such as dropdowns, sliders, tabs, and data tables. These components allow users to filter results, view transport details, and navigate through different modules efficiently. Python is the core programming language used to implement the application logic. It handles user input processing, voice recognition integration, data manipulation, and communication with the database.

CHAPTER 2

LITERATURE SURVEY

2.1 REQUIREMENT ANALYSIS

The The first step in developing the Voice-Based Transport Enquiry System is to identify both functional and non-functional requirements. These requirements define what the system should perform and the conditions under which it should operate efficiently.

Functional Requirements:

- User Registration and Login
- Voice-Based Enquiry for transport search
- Manual Enquiry using source and destination selection
- Displaying transport details such as fare, timing, and vehicle type
- Admin functionalities: add, update, and delete transport records, manage users
- Route visualization and result filtering options

Non-Functional Requirements:

- Usability: Simple, interactive, and user-friendly interface using Streamlit
- Performance: Fast query execution and efficient data retrieval from the database
- Security: Password hashing, authentication control, and secure database operations
- Scalability: Easily extendable to include features like real-time tracking and booking
- Maintainability: Modular code structure with clear separation of functionalities

2.2 ENTITY-RELATIONSHIP (ER) DIAGRAM

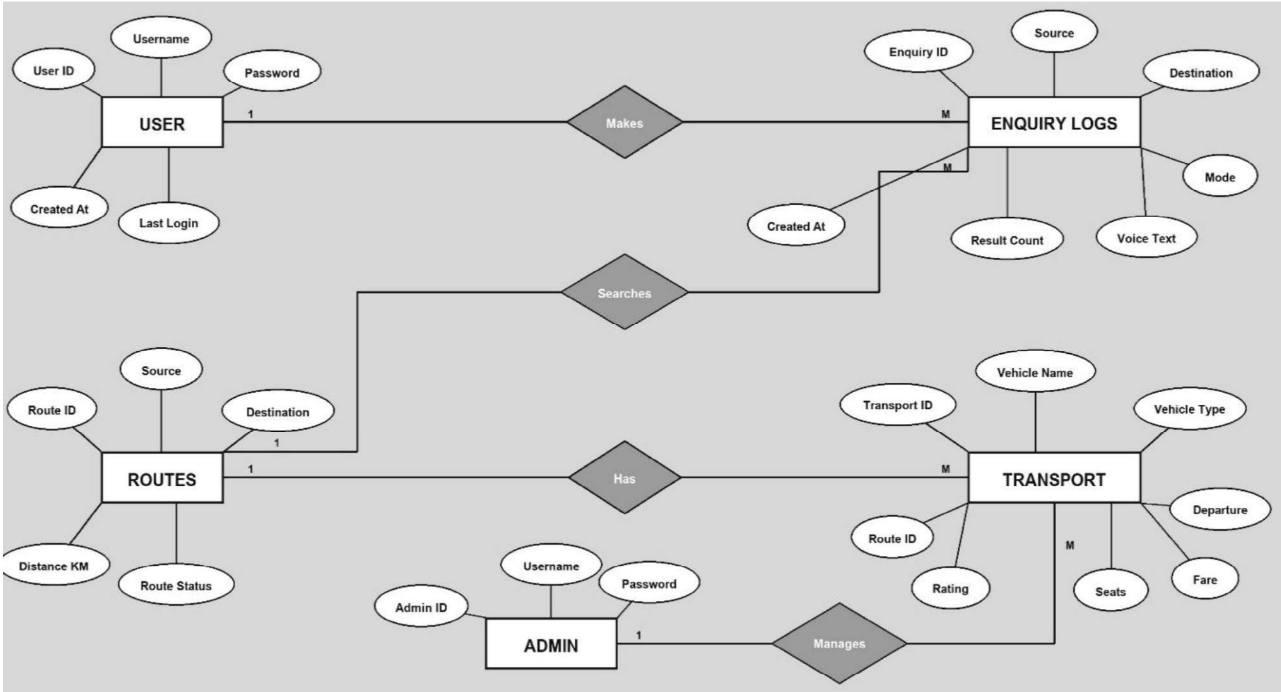

Figure 2.2.1 Entity Relationship

The ER Diagram illustrates how different entities in the Voice-Based Transport Enquiry System are interconnected. It provides a structured representation of the database design and defines how data is stored, accessed, and related across different modules.

Main Entities:

- **User:** Stores user details such as user ID, username, password, created date, and last login information.
- **Routes:** Contains route ID, source, destination, distance, and route status, representing available travel paths between cities.
- **Transport:** Includes transport ID, vehicle name, vehicle type, route ID (foreign key), departure time, fare, seats available, and service rating.
- **Enquiry_Logs:** Stores enquiry ID, source, destination, enquiry mode (voice/manual), voice text, result count, and timestamp of the enquiry.

- **Admin:** Represents administrative users with privileges to manage transport records and system data.

Relationships:

- One user can make multiple enquiries (1-to-many relationship between User and Enquiry_Logs).
- One enquiry searches for a specific route (relationship between Enquiry_Logs and Routes).
- One route can have multiple transport options (1-to-many relationship between Routes and Transport).
- One admin can manage multiple transport records (1-to-many relationship between Admin and Transport).

2.3 RELATIONAL SCHEMA

Based on the ER diagram, the relational schema defines how data is organized into tables and how they are connected using primary and foreign keys.

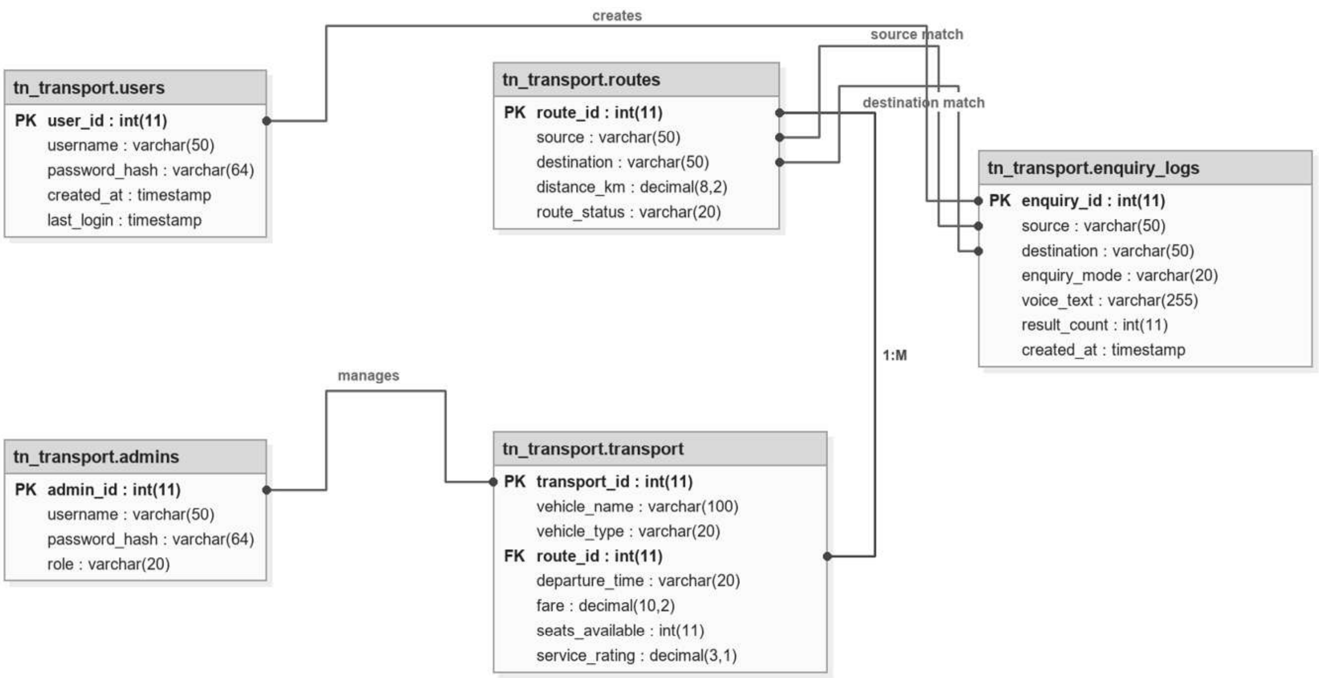

Figure 2.3.1 Relational Schema

Schema Example:

• **Users(User_ID, Username, Password_Hash, Created_At, Last_Login)**

→ User_ID is the Primary Key

• **Routes(Route_ID, Source, Destination, Distance_KM, Route_Status)**

→ Route_ID is the Primary Key

• **Transport(Transport_ID, Vehicle_Name, Vehicle_Type, Route_ID, Fare, Departure_Time, Seats_Available, Service_Rating)**

→ Transport_ID is the Primary Key, Route_ID is a Foreign Key referencing Routes

• **Enquiry_Logs(Enquiry_ID, Source, Destination, Enquiry_Mode, Voice_Text, Result_Count, Created_At)**

→ Enquiry_ID is the Primary Key

• **Admins(Admin_ID, Username, Password_Hash, Role)**

→ Admin_ID is the Primary Key

The relationships between these tables are maintained using foreign keys, such as Route_ID in the Transport table linking it to the Routes table. This structure ensures proper data organization, avoids redundancy, and supports efficient querying of transport and enquiry data. This schema is normalized to maintain data consistency, integrity, and scalability within the system.

2.4 NORMALIZATION PROCESS

Normalization is the process of organizing data in the database to minimize redundancy and ensure efficient data access. The **Voice-Based Transport Enquiry System** applies the first three normal forms to maintain a well-structured database design.

• **1NF (First Normal Form):** In this system, attributes such as source, destination, fare, and vehicle details are stored as individual values without repeating groups.

• **2NF (Second Normal Form):** Attributes like vehicle name, fare, and departure time depend entirely on the transport_id, eliminating partial dependency.

• **3NF (Third Normal Form):** In this system, route details such as source and destination are stored separately in the routes table and linked using route_id, avoiding duplication in the transport table.

2.5 SYSTEM ARCHITECTURE OVERVIEW

The architecture of the Voice-Based Transport Enquiry System follows a three-tier model:

1. Presentation Layer (Frontend): Built using Streamlit

2. Application Layer (Backend): Developed using Python, handles core functions of the system

3. Data Layer (Database): Managed using MySQL, this layer stores all persistent data including user accounts, transport records, routes, and enquiry logs.

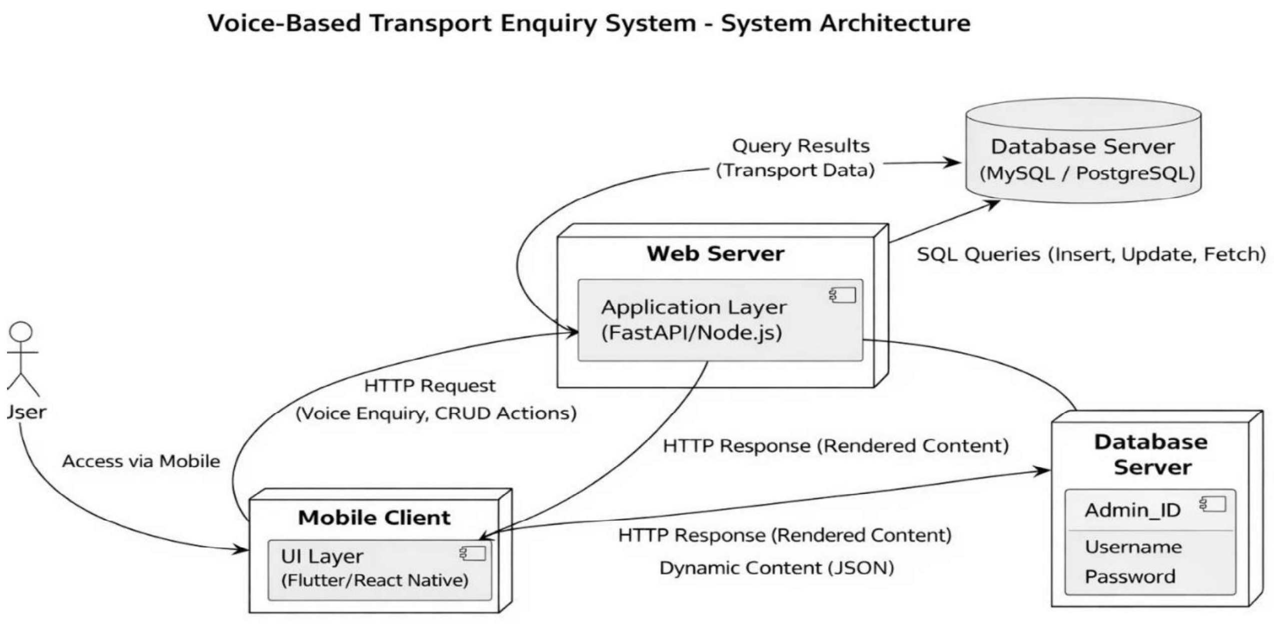

Figure 2.5.1 System Architecture

The methodology adopted for the Voice-Based Transport Enquiry System ensures a balance between functionality, performance. Through proper requirement analysis, efficient database design, normalization, and layered architecture, the project establishes a strong foundation and intelligent transport enquiry application.

CHAPTER 3

PROJECT METHODOLOGY

3.1 VOICE ENQUIRY MODULE

The Voice Enquiry Module handles all functionalities related to voice-based transport search, including capturing speech input, processing audio, extracting relevant information, and retrieving results from the database. It plays a vital role in enabling hands-free interaction, ensuring that users can access transport details quickly and efficiently using natural language.

Key Features:

- **Voice Input Capture:** Allows users to provide input through a microphone. The system listens to the user's speech and records the query for further processing.
- **Speech-to-Text Conversion:** Converts spoken input into text using a speech recognition library, enabling the system to interpret user queries.
- **City Extraction and Correction:** Identifies source and destination cities from the recognized text and applies correction techniques to handle mispronunciations or variations.
- **Voice Response Output:** Generates audio responses by converting the result summary into speech, improving accessibility and user experience.

Functionality Flow:

1. User activates the voice input feature and speaks the transport query.
2. The system captures audio and converts it into text using speech recognition.
3. Extracted text is processed to identify valid source and destination cities.
4. The system queries the database and displays results, along with voice output.

3.2 MANUAL ENQUIRY MODULE

This is the core functional module of the system. It allows users to search for transport details by manually selecting source and destination locations. It supports displaying transport results in structured formats along with filtering and visualization features.

Key Features:

- **Transport Search:** Authenticated users can select source and destination cities using dropdown menus in a user-friendly interface. Based on the selection, the system retrieves matching transport records such as vehicle name, type, fare, and departure time.
- **Display Results:** All transport options are fetched from the database and displayed using tables and smart cards, making it easy for users to compare available choices.
- **Route Visualization:** The module provides a live route map with distance and time estimation, enhancing user understanding of the journey.

Functionality Flow:

1. User selects source and destination cities from the interface.
2. The system validates the input and sends a query to the database.
3. Matching transport records are retrieved and displayed to the user.
4. Users can apply filters or modify search criteria to refine results.

3.3 DBMS ANALYTICS MODULE

This module focuses on analyzing stored data in the system to generate meaningful insights about transport usage and user enquiry patterns. It utilizes database queries to process information from transport records and enquiry logs, helping in understanding system performance and user behavior.

Key Features:

• **Data Analysis:** Retrieves and processes data such as total routes, number of transport records, and average fare from the database.

• **Enquiry Mode Statistics:** Compares usage between voice-based and manual enquiries using grouped database queries.

• **Popular Routes Identification:** Identifies frequently searched source-to-destination routes based on enquiry logs.

• **Visual Representation:** Displays results using charts and tables, making it easier to interpret trends and patterns.

Functionality Flow:

1. The system retrieves aggregated data from transport and enquiry logs tables.
2. Database queries are executed to calculate statistics such as counts and averages.
3. Results are processed and formatted for visualization.

3.4 ADMIN CONTROL & MANAGEMENT MODULE

This module provides administrative control over the system, allowing authorized users to manage transport data and user accounts. It ensures that the database remains accurate, updated, and secure.

Key Features:

• **Transport Management:** Admins can add new transport records, update existing entries, and delete outdated data from the system.

• **User Management:** Allows administrators to view registered users, update user details, and delete accounts when necessary.

• **Data Monitoring:** Provides access to all transport and user data for effective system management.

• **Access Control:** Ensures that only authorized admin users can perform management operations.

Functionality Flow:

1. Admin logs into the system with valid credentials.
2. Admin accesses the dashboard to view transport and user data.
3. Admin performs operations such as adding, updating, or deleting records.
4. Changes are saved in the database and reflected in the system immediately.

CHAPTER 4

DATABASE IMPLEMENTATION

A robust and well-structured database is the backbone of the **Voice-Based Transport Enquiry System**. In this project, the database plays a crucial role in storing, organizing, and retrieving transport-related data such as user accounts, transport records, routes, and enquiry logs. This chapter explains how the database is designed and implemented, including table structures, relationships, constraints.

4.1 TABLE CREATION SCRIPTS

- The system uses a relational database management system (RDBMS), specifically MySQL, to store structured transport and user data. The following tables form the core of the database:

1. Users Table

```
sql
Copy code
CREATE TABLE users (
    user_id INT PRIMARY KEY AUTO_INCREMENT,
    username VARCHAR(50) NOT NULL UNIQUE,
    password_hash VARCHAR(64) NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    last_login TIMESTAMP NULL DEFAULT NULL
);
```

- **Purpose:** Stores information about all registered users, including authentication details and login activity.

2. Routes Table

```
sql
Copy code
CREATE TABLE routes (
    route_id INT PRIMARY KEY AUTO_INCREMENT,
    source VARCHAR(50) NOT NULL,
    destination VARCHAR(50) NOT NULL,
    distance_km DECIMAL(8,2),
    route_status VARCHAR(20)
);
```

- **Purpose:** Stores route information including source and destination cities along with distance and route status.

3. Transport Table

```
sql
Copy code
CREATE TABLE transport (
    transport_id INT PRIMARY KEY AUTO_INCREMENT,
    vehicle_name VARCHAR(100),
    vehicle_type VARCHAR(20),
    route_id INT,
    departure_time VARCHAR(20),
    fare DECIMAL(10,2),
    seats_available INT,
    service_rating DECIMAL(3,1),
    FOREIGN KEY (route_id) REFERENCES routes(route_id)
);
```

- **Purpose:** Stores transport details such as vehicle type, fare, timing, availability, linked to routes.

4. Enquiry_Logs Table

```
sql
Copy code
CREATE TABLE enquiry_logs (
    enquiry_id INT PRIMARY KEY AUTO_INCREMENT,
    source VARCHAR(50) NOT NULL,
    destination VARCHAR(50) NOT NULL,
    enquiry_mode VARCHAR(20) NOT NULL,
    voice_text VARCHAR(255),
    result_count INT DEFAULT 0,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

- **Purpose:** Stores all user enquiries like source, destination, mode of enquiry.

5. Admins Table

```
sql
Copy code
CREATE TABLE admins (
    admin_id INT PRIMARY KEY
  AUTO_INCREMENT,
    username VARCHAR(50),
    password_hash
  VARCHAR(64),
    role VARCHAR(20)
);
```

- **Purpose:** Stores admin credentials and roles for managing transport records and user data.

4.2 CONSTRAINTS AND RELATIONSHIPS

Implementing **primary keys**, **foreign keys**, and **constraints** ensures that the database maintains data integrity, avoids redundancy in Voice-Based Transport Enquiry System. Primary keys uniquely identify each record in a table (e.g., user_id, route_id, transport_id, enquiry_id, admin_id). These keys ensure that each entry in the database is distinct and can be accessed efficiently.

Foreign keys are used to establish relationships between tables. For example, route_id in the Transport table links to the Routes table. These relationships help maintain consistency and reflect real-world connections between routes and transport services.

Constraints such as NOT NULL Constraint: Ensures that important fields such as username, password, source, and destination are always provided and cannot be left empty. UNIQUE Constraint: Applied to fields like username to prevent duplicate user registrations in the system. DEFAULT Constraint: Automatically assigns default values such as timestamps (created_at) and result_count when data is inserted.

4.3 BACKUP & RECOVERY CONSIDERATIONS

Ensuring data safety and reliability is essential for the Voice-Based Transport Enquiry System. The database includes backup and recovery strategies to prevent data loss due to system failures. Regular database backups can be scheduled using tools like mysqldump to create full database dumps of users, transport, routes, and enquiry logs tables. Data Export Options: The system allows exporting enquiry results and logs in formats like CSV, enabling offline storage and manual backup of important data. Recovery Strategies include Point-in-Time Recovery: Using MySQL logs, the database can be restored to a specific state before data loss or corruption occurred. Manual Restoration: Backup files can be reloaded into the database to restore lost data such as transport records or enquiry history. Future Enhancements: Advanced recovery features like version tracking of transport data or automated rollback systems can be implemented in future versions.

CHAPTER 5
RESULTS AND DISCUSSION

After the successful development and implementation of the Voice-Based Transport Enquiry System, the application was thoroughly tested to evaluate its functionality, performance, and user experience. This chapter presents the results of these evaluations and discusses how effectively the system meets its objectives, along with areas for further improvement.

5.1 PERFORMANCE ANALYSIS

The performance of the Voice-Based Transport Enquiry System was evaluated across key components, including response time, database operations, user interface responsiveness, and overall system reliability under normal usage conditions.

Frontend Performance: The Streamlit-based interface was responsive and functioned smoothly across different screen sizes, including desktops and laptops. Page loading and interaction times were minimal, with most operations completing within a few seconds during local execution. Navigation between modules such as voice enquiry, manual enquiry, analytics, and admin panel was seamless, providing a user-friendly experience.

Backend Processing: Voice input processing using speech recognition worked effectively, accurately converting speech into text and extracting source and destination cities. Manual enquiry processing and filtering operations were executed efficiently without delays. Core operations such as fetching transport data, logging enquiries, and handling admin actions were performed reliably without data inconsistency.

Database Operations: SQL queries for retrieving transport records and routes returned results quickly due to structured schema design and optimized queries. Insert operations for enquiry logs and updates to transport records were handled efficiently and consistently. Relationships between tables such as routes and transport ensured accurate data retrieval and integrity.

Security Measures: User passwords were securely stored using hashing techniques, preventing direct exposure of sensitive data. Input validation and controlled query execution helped prevent invalid data entry and potential database issues. Role-based access ensured that only authorized users could access admin functionalities.

Scalability (Initial Observation): The system handled multiple transport records and enquiry logs without noticeable performance degradation during testing. The architecture supports future scalability through enhancements such as indexing, caching, and integration with real-time APIs. Modular design allows easy addition of new features like live tracking, booking systems, and multilingual voice support.

5.2 OBSERVATIONS AND INTERPRETATIONS

The system was intuitive and easy to navigate, with clear modules such as voice enquiry, manual enquiry, analytics, and admin panel. Users appreciated the voice-based enquiry feature, which allowed them to search transport details quickly without manual input. The combination of visual results (cards, tables) and route maps enhanced understanding and improved overall usability. Challenges Observed: Voice recognition occasionally faced issues in noisy environments or with unclear pronunciation, affecting accuracy. Users sometimes needed to repeat queries if correct source and destination cities were not detected.

Admin Insights: Admin users were able to efficiently manage transport records, including adding, updating, and deleting entries. The analytics module helped in identifying popular routes and system usage patterns, though more advanced visual dashboards could further improve insights.

Functional Stability: All major modules (voice enquiry, manual enquiry, DBMS analytics, and admin control) worked reliably without crashes during testing. Database operations such as fetching transport data and logging enquiries were executed consistently. Visual and Layout Aspects: The UI was clean and minimalistic, allowing users to focus on content. Use of responsive design ensured accessibility across various screen sizes.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 SUMMARY OF OUTCOMES

The performance of the Voice-Based Transport Enquiry System was evaluated based on three main criteria: response time, system functionality, and overall user experience through the web interface. Voice Enquiry Module: The speech recognition system successfully converted voice input into text and accurately extracted source and destination cities in most cases. The integration of voice input with database queries provided a fast and hands-free enquiry experience. Database & Processing Efficiency: Optimized SQL queries and structured database design enabled quick retrieval of transport records, ensuring minimal delay in displaying results. Web Interface: The Streamlit-based interface provided smooth interaction, with most operations such as search, filtering, and result display completing within a few seconds. The integration of maps and voice output enhanced usability. Overall System: The system achieved a good balance between functionality and usability, combining voice interaction, manual search, analytics, and admin control into a single platform. Limitations: Voice recognition accuracy may decrease in noisy environments or due to unclear pronunciation, affecting query results. The system currently depends on predefined routes and database records, limiting real-time transport updates. As the number of users and data increases, performance may be affected unless database optimization and scaling techniques are implemented.

6.2 FUTURE SCOPE AND ENHANCEMENTS

While the core features of the Voice-Based Transport Enquiry System are successfully implemented, there is significant scope for future improvements and expansion. These enhancements can transform the system from a basic enquiry platform into a comprehensive intelligent transport solution. Multilingual Voice Support: Integration of multiple languages such as Tamil and English to improve accessibility for a wider range of users. Real-Time Transport Data: Integration with live transport APIs to provide real-time updates on bus/train schedules, delays, and availability. Advanced Search and Filtering: Implement more refined search options such as time-based filtering, preferred transport type, and shortest/fastest route selection. Booking Integration: Allow users to book tickets directly through the system after viewing transport options. User Profile Management: Provide user dashboards to track previous enquiries, preferences, and personalized suggestions. AI-Based Recommendations: Use machine learning algorithms to suggest optimal routes based on user behavior, traffic conditions, historical data. Enhanced Analytics Dashboard: Expand admin analytics to include detailed insights such as peak usage times, user trends, and route demand patterns. Mobile Application: Develop a mobile version of the system using frameworks like Flutter or React Native for improved accessibility and convenience. Improved Voice Accuracy: Incorporate advanced NLP and speech models to handle complex queries and reduce errors in voice recognition.

In conclusion, this project not only delivers a functional Voice-Based Transport Enquiry System but also establishes a strong foundation for future enhancements. It highlights the importance of modular design, database integration, and intelligent user interaction in building scalable and practical applications.

APPENDICES

APPENDIX A

SOURCE CODE

This section presents key source code snippets from the Voice-Based Transport Enquiry System application. The code is written using Python for backend processing and MySQL for the database. The source code follows a modular and secure approach, with proper validation, authentication, and database interaction.

A.1 User Registration Module

Purpose: Allows new users to register and securely stores their credentials in the database.

Python

```
def register_user(username, password):
    if run_query("SELECT user_id
FROM users WHERE username = %s",
(username,)):
        return False, "Username already
exists."
    success = run_query(
        "INSERT INTO users (username,
password_hash) VALUES (%s, %s)",
        (username,
hash_password(password)),
        fetch=False,
    )
    return (True, "Registration
successful.") if success else (False,
"Registration failed.")
```

Explanation:

- Accepts username and password from the form.
- Password is hashed using SHA-256 for security.
- Data is stored in the Users table.

A.2 User Login Module

Purpose: Authenticates users and starts a session for logged-in users.

Python

```
def authenticate_user(username, password):
    result = run_query(
        "SELECT user_id FROM users WHERE username=%s AND
password_hash=%s",
        (username, hash_password(password))
    )
    return result
```

Explanation:

- Fetches user record based on username.
- Verifies password using hashed comparison.
- Allows access if credentials are valid.

A.3 Voice Enquiry Module

Purpose: Enables users to search transport details using voice input.

Python

```
def extract_cities_from_text(text, supported_cities):
    words = text.split()
```

```
if "from" in words and "to" in words:
        source = words[words.index("from")+1]
        destination = words[words.index("to")+1]
        return source, destination
return None, None
```

Explanation:

- Captures voice input and converts it to text.
- Extracts source and destination cities.
- Used for querying transport data.

A.4 Manual Enquiry Module

Purpose: Allows users to search transport details manually.

Python

```python
def fetch_search_results(src, dst):
    return run_query(
        "SELECT * FROM transport t JOIN routes r ON t.route_id=r.route_id
WHERE r.source=%s AND r.destination=%s",
        (src, dst)
    )
```

Explanation:

- Accepts source and destination from UI.
- Fetches matching transport data from database.

A.5 Display Results Module

Purpose: Displays transport details and route visualization.

Python

```
route = get_route_details(src, dst)
render_route_map(route, src, dst)
```

Explanation:

- Retrieves route details using API.
- Displays map with distance and duration.
- Enhances user understanding visually.

A.6 Admin Management Module

Purpose: Allows admin to manage users and transport data.

Python

```
def delete_user_account(user_id):
    return run_query("DELETE FROM users WHERE user_id=%s", (user_id,),
fetch=False)
```

Explanation:

- Allows admin to delete users.
- Manages system data securely.
- Ensures controlled access.

The source code demonstrates key modules that contribute to the functioning of the Voice-Based Transport Enquiry System. These include secure authentication, voice and manual enquiry processing, data visualization, and admin controls.

APPENDIX B – Screenshots

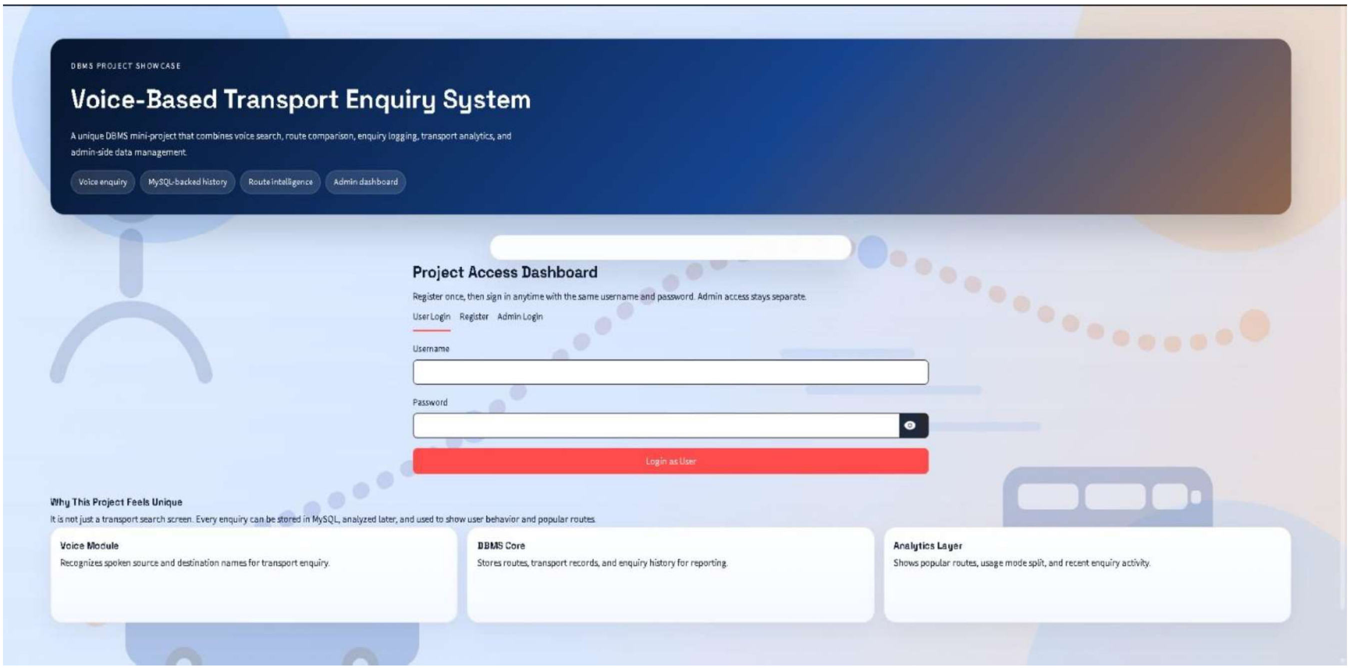

Figure B.1 LOGIN PAGE

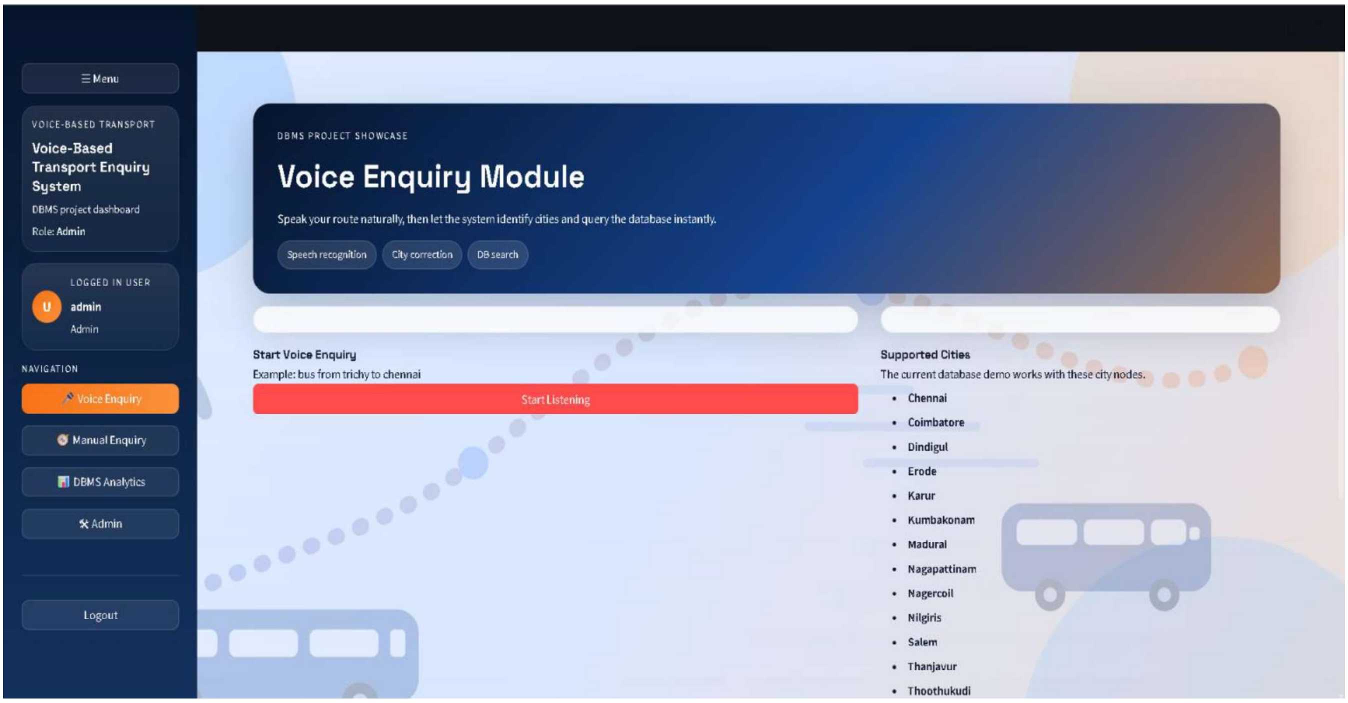

Figure B.2 VOICE ENQUIRY MODULE

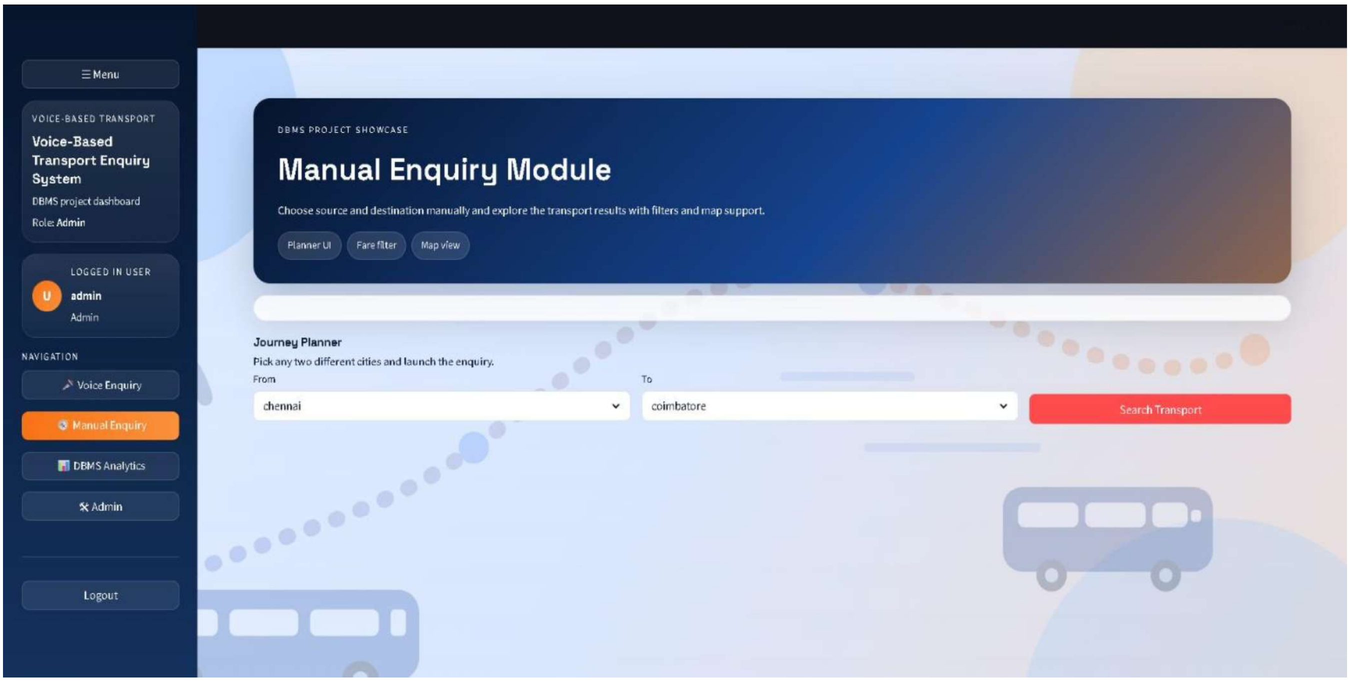

Figure B.3 MANUAL ENQUIRY MODULE

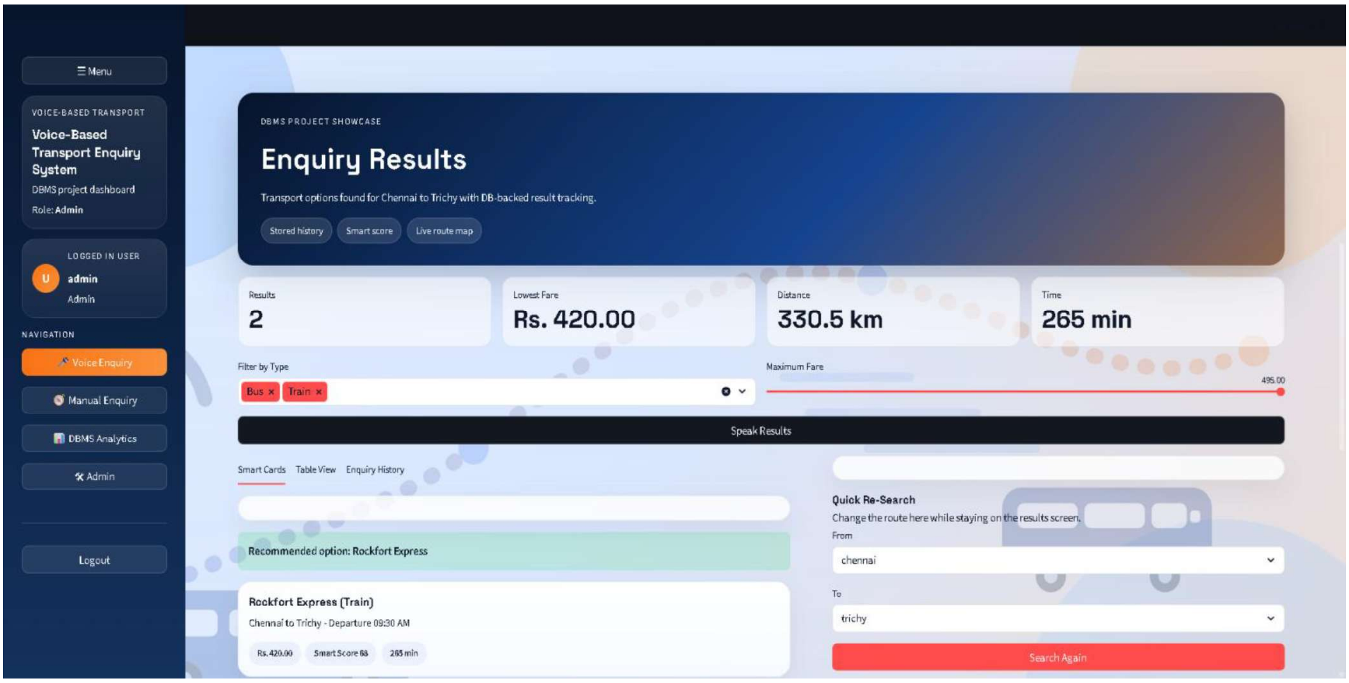

Figure B.4 DISPLAY RESULTS MODULE

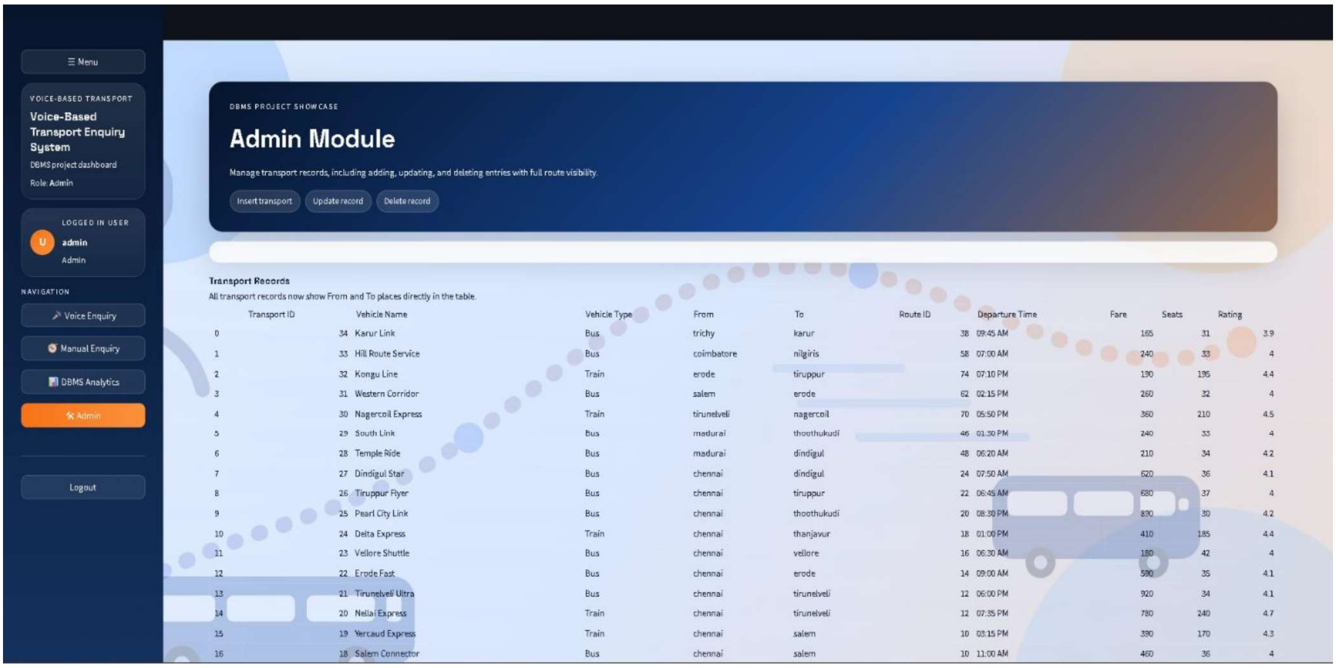

Figure B.5A ADMIN MANAGEMENT MODULE

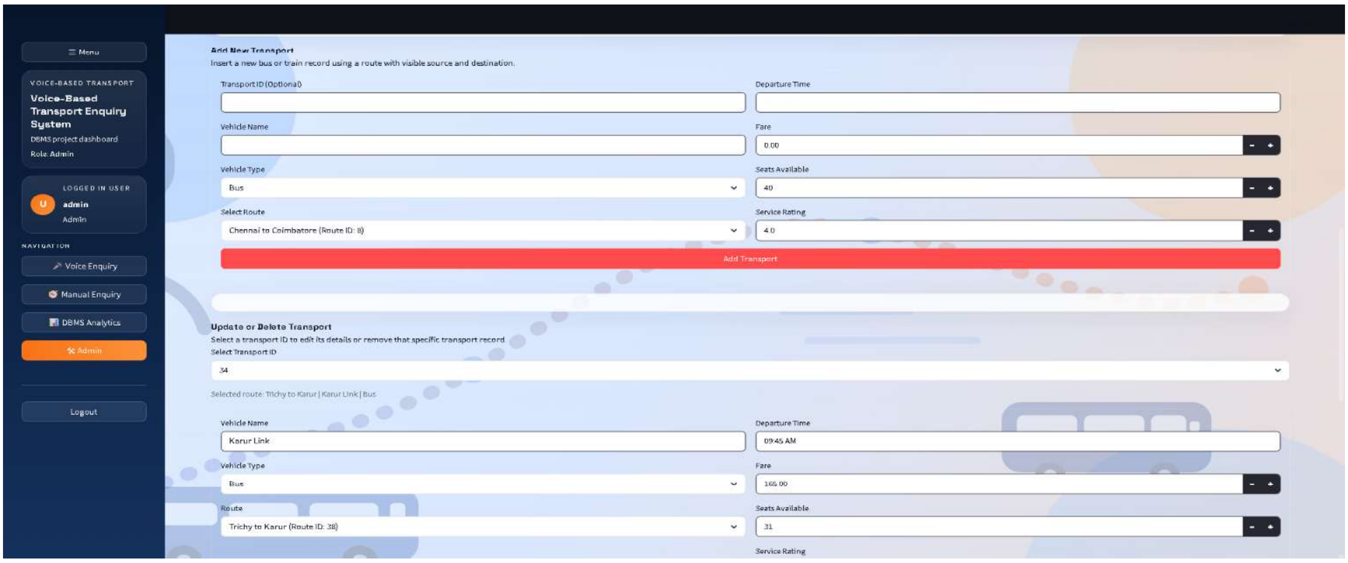

Figure B.5B ADMIN MANAGEMENT MODULE

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